

THE SCHOOL OF COMPUTER SCIENCES

Bat Face Recognition Using Deep Learning

Research Summary Jul-2025

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1 Introduction

Accurate identification of individual animals from visual data has significant implications for behavioral research and wildlife monitoring. Traditionally, species such as bats have been identified through tagging or manual marking—methods that are invasive, prone to human error, and unsuitable for long-term behavioral studies.

In this work, we explore the evolutionary underpinnings of visual facial recognition in bats. We hypothesize that species active during the day, such as *Taphozous mauritanus* (Mauritius tomb bats), have developed distinct facial characteristics to enable individual recognition within well-lit environments. In contrast, nocturnal species like *Rousettus aegyptiacus*, which rely less on vision, are not expected to exhibit the same level of facial differentiation, having evolved other sensory modalities instead.

To test these hypotheses, we applied a Siamese neural network—a state-of-the-art deep learning architecture for measuring similarity between pairs of inputs—to learn face embeddings. Underlying this approach is the assumption that the network intrinsically captures the most discriminative features that distinguish individuals.

We designed the study to test two core hypotheses:

1. The Siamese network will perform better at distinguishing *Mauritius* bats compared to *Rousettus*, reflecting visual differentiation evolution.
2. Saliency map analysis will reveal that the network focuses on specific facial features (e.g., eyes, nose, mouth) in *Mauritius* bats, but relies more on general face shape in *Rousettus*.

Our results support these expectations: first, the Siamese model achieves significantly higher accuracy on *Mauritius* bats; second, aggregate saliency visualization indicates that discrimination in *Mauritius* is driven by key facial landmarks, while for *Rousettus*, the network focuses on more global structural cues.

2 Training Dataset Overview and Processing Pipeline

This section outlines the process used to prepare the data for training the bat face recognition system, highlighting differences in data collection conditions between species and describing the complete image processing pipeline.

2.1 Training Data Collection Conditions

Mauritius bats: The raw data for Mauritius bats were extracted from video recordings captured under varied lighting conditions, times of day, and dates. This introduced natural variability in the environmental context surrounding each individual, simulating real-world diversity.

Rousettus bats: In contrast, the Rousettus dataset was captured manually as static images by the researcher. All images were taken in a controlled environment with identical lighting and

background conditions for all individuals. This consistency minimizes background-induced variation across the dataset.

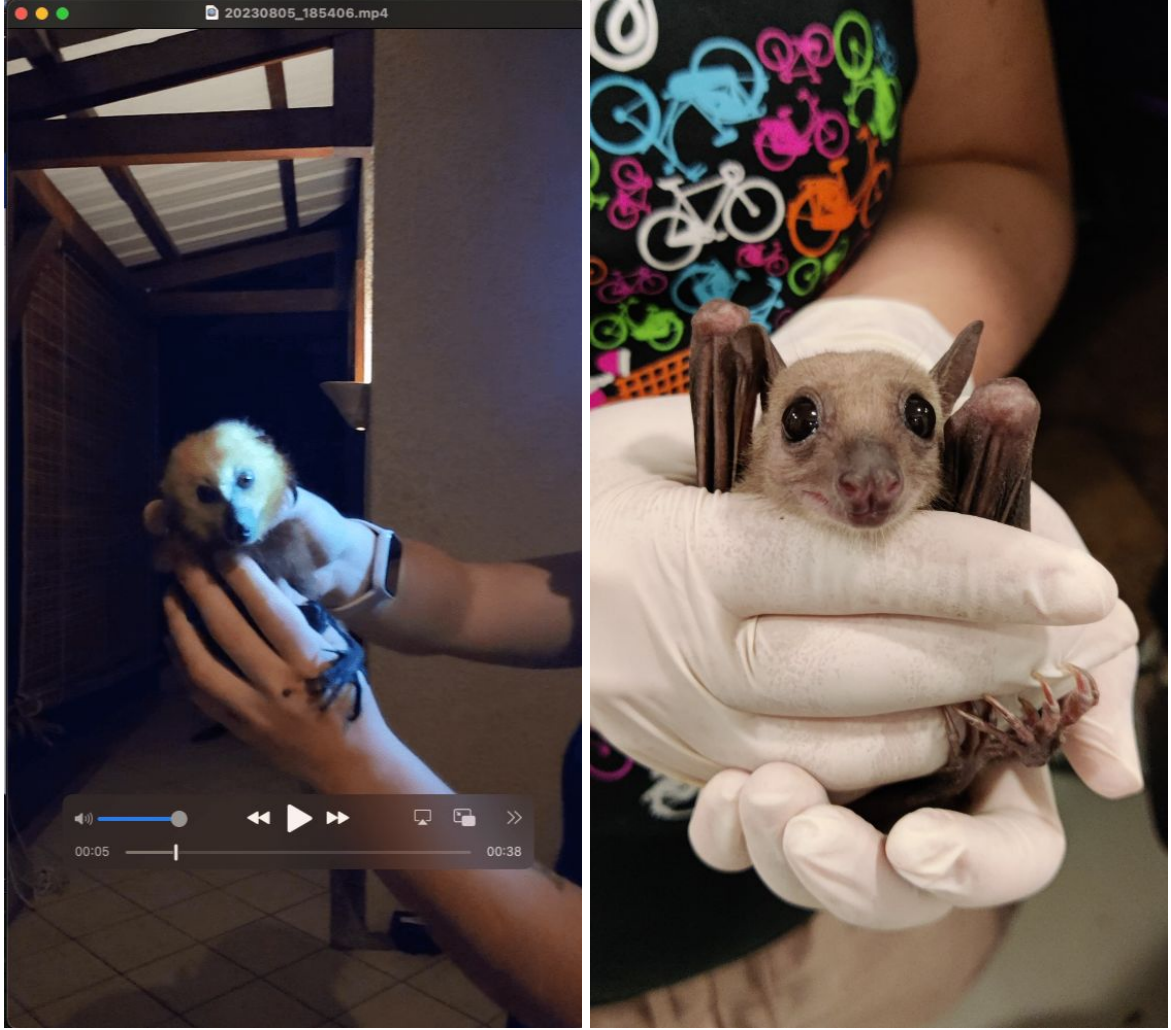


Figure 1: Left: Sample Mauritius video frame (variable environment). Right: Rousettus image (controlled environment).

2.2 Data Processing Pipeline

Depending on the experiment, a subset of the following steps was applied to each image:

1. **Face Snapshotting:** Extracted face frames from video sources using a custom-trained YOLOv8 detection model.
2. **Frontal Filtering:** Retained only images presenting a frontal view of the bat.
3. **Augmentation:** Applied up to 60 variants per frame using brightness, rotation, contrast, color, and scaling augmentations.
4. **Background Replacement:** Removed original background and substituted using segmentation masks from a custom YOLOv8 segmentation model.

5. **Cropping and Reshaping:** Normalized image size and aspect ratio to fit Siamese network input dimensions.
6. **Siamese Evaluation:** Processed final image pairs through a Siamese network to measure facial similarity.

2.3 Pipeline per Species

2.3.1 Mauritius Bats

All experiments on Mauritius bats began with face snapshots taken from video data using a custom-trained YOLOv8 model. These videos were collected under various conditions: different lighting environments, times of day, and recording dates, resulting in a naturally diverse dataset.

2.3.1.1 Multi-Angle with Augmentation This experiment used both frontal and nonfrontal images. Each detected face was augmented up to 60 times using a wide range of transformations such as brightness shifts, rotations, contrast tuning, and geometric scaling. This approach emphasized dataset expansion to test if variety improves identity recognition.



2.3.1.2 Frontal Faces Only Only images showing a clear frontal view of the bats were included.



2.3.1.3 Green Background Replacement Frontal images were selected, then processed to remove background using segmentation masks. The background was replaced with a solid green screen to

neutralize contextual elements such as hands or lighting inconsistencies.



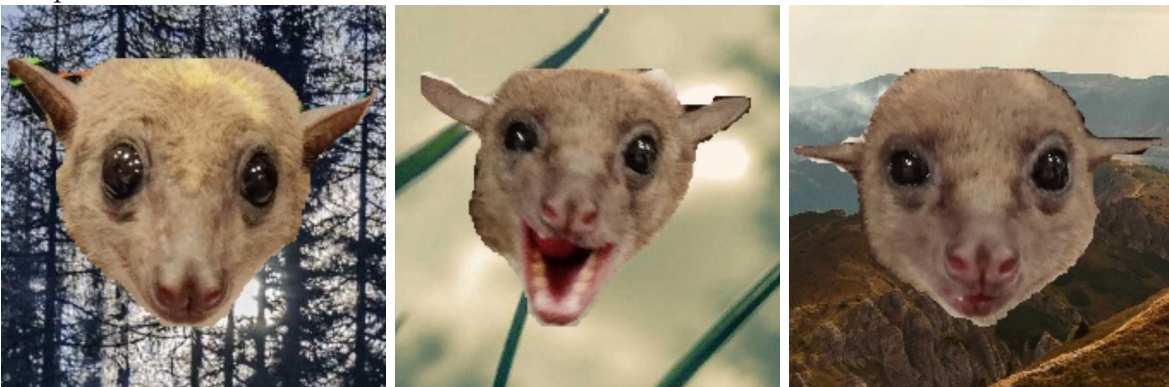
2.3.1.4 Random Natural Backgrounds Building on the front-only set, this experiment replaced the background of each image with a unique natural scene sourced from the Picsum API. The aim was to add rich but noninformative variation to help the network ignore background cues.



All pipelines ended with cropping, reshaping, and a final pairwise evaluation using the Siamese network.

2.3.2 Rousettus Bats

Images underwent frontal face filtering, their backgrounds were also replaced. As with the Mauritius bats, all inputs were finally reshaped and passed through the Siamese network for identity comparison.



3 Results Overview

After conducting multiple training iterations with Mauritius bats, we found that the version which utilized only frontal face images and applied randomized natural background replacement provided the most robust and interpretable results. This configuration not only performed well in classification accuracy, but also demonstrated reduced reliance on environmental artifacts such as background textures, lighting direction, or the presence of human hands. It enabled the model to focus on actual facial features, making the network more biologically aligned with the hypothesis of sight-based recognition.

3.1 Mauritius Bat Results

The final Mauritius model achieved the best results using frontal-only images with random natural backgrounds. Saliency maps from this version consistently focused on key anatomical features including the eyes, snout, and mouth. These cues align with the hypothesis that Mauritius bats have evolved visually distinct facial features to facilitate individual identification in daylight environments. The classification accuracy in this setting reached approximately 70%, and generalization performance held up across different recordings and lighting conditions.

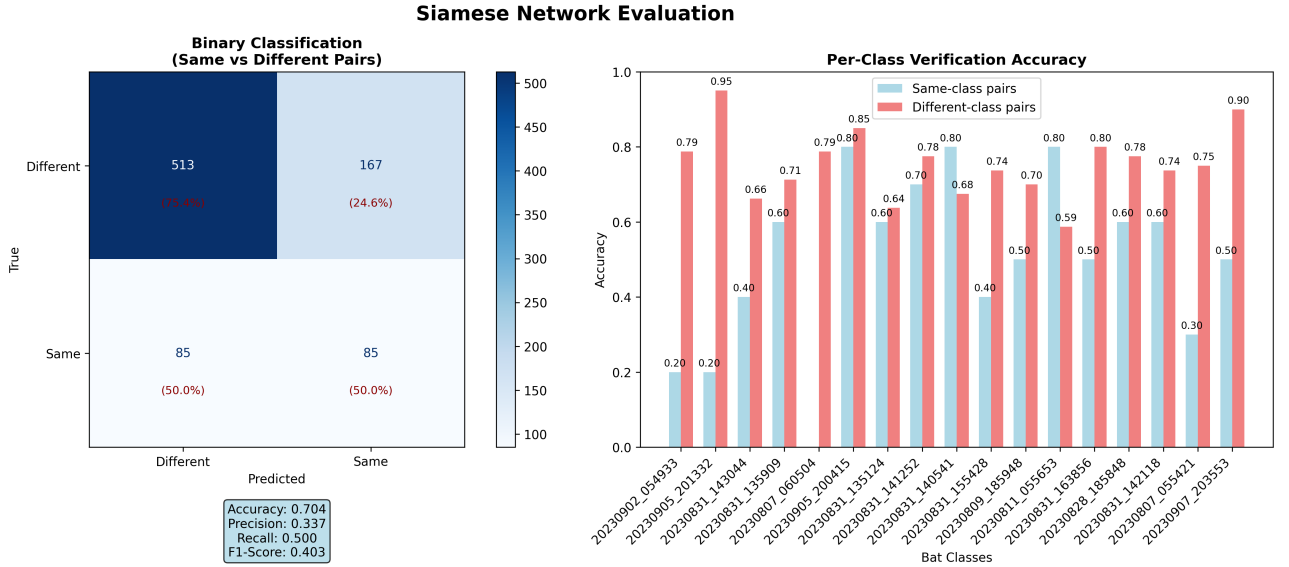


Figure 2: Confusion Matrix for Mauritius on unprocessed data of bats the model have seen.

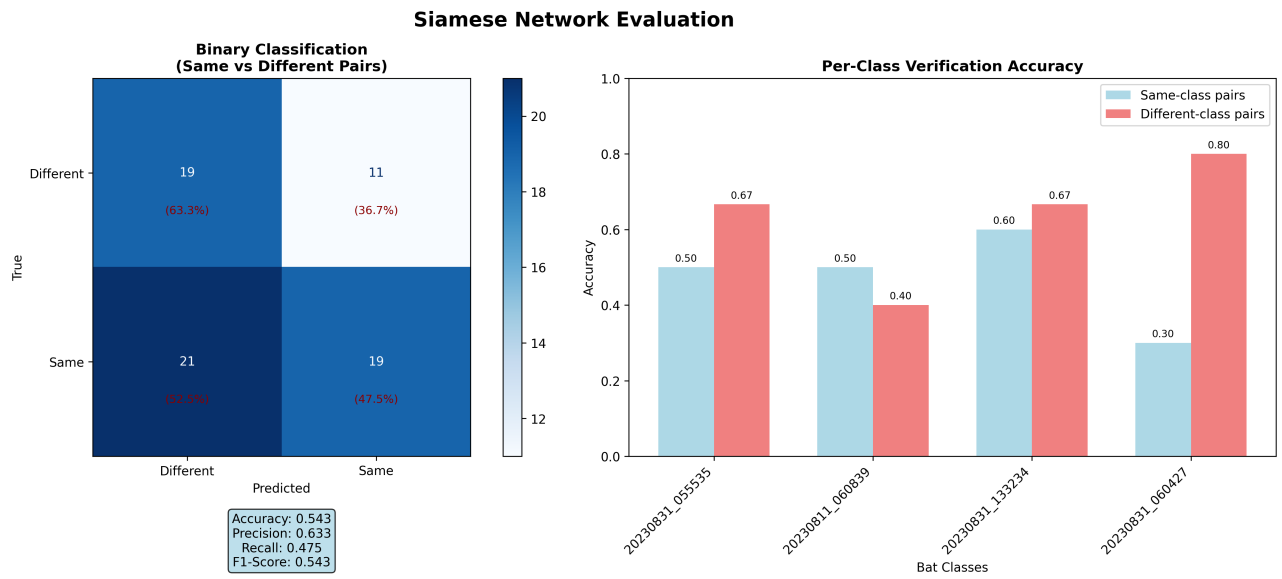


Figure 3: Confusion Matrix for Mauritius on unprocessed data of bats the model haven't seen.

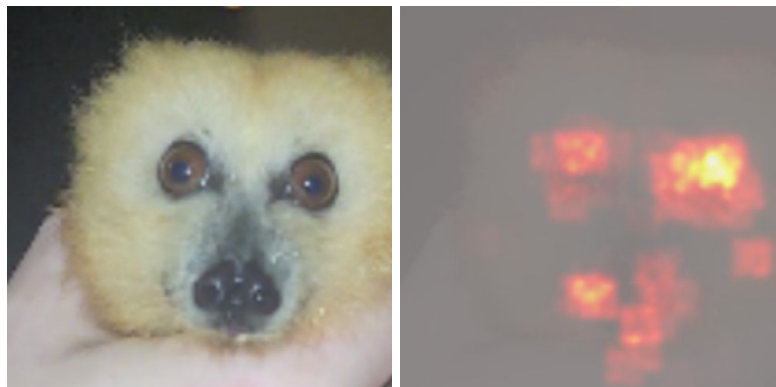


Figure 4: Original image (Left) vs. its matching Saliency Heatmap (Right).

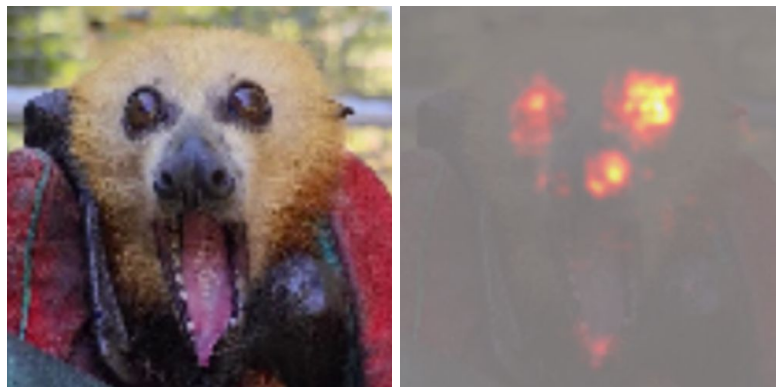


Figure 5: The Original image (Left) vs. its matching Saliency Heatmap (Right).

3.2 Rousettus Bat Results

For the Rousettus dataset, which included uniformly captured images under constant conditions, the network achieved moderate accuracy. However, saliency visualization revealed that the network’s focus was more diffused and often distributed along the entire facial outline. This suggests that unlike Mauritius bats, Rousettus individuals may not have evolved to exhibit highly differentiable facial characteristics—consistent with their nocturnal behavior and reduced reliance on visual identification. While some classification was achieved, the network’s interpretability and confidence were lower compared to the Mauritius case.

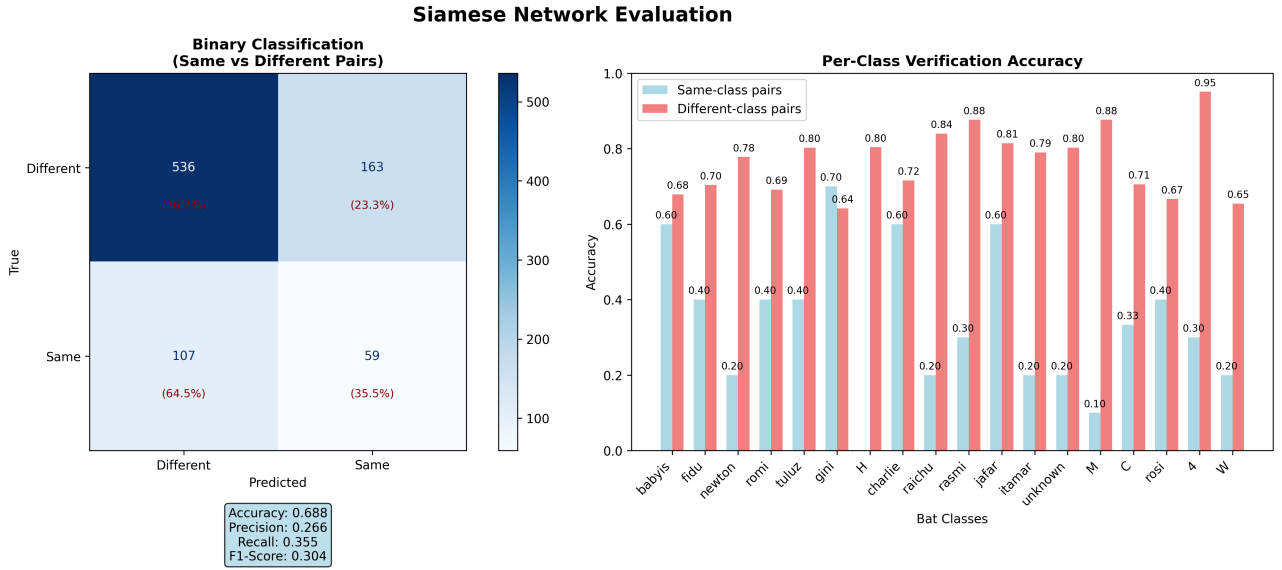


Figure 6: Confusion Matrix for Rousettus on unprocessed data of bats the model have seen.

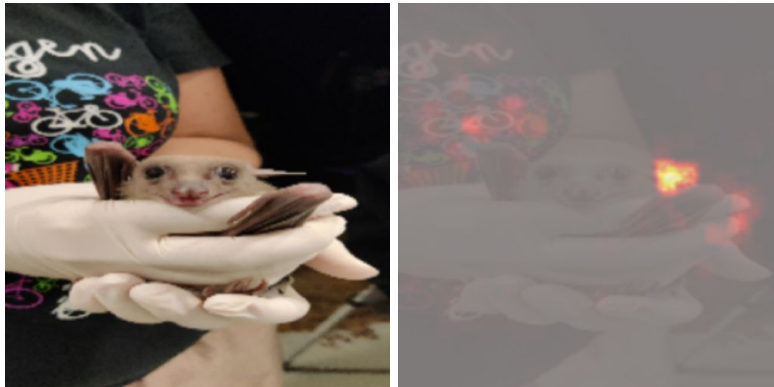


Figure 7: The bat "Babyis". Original image (Left) vs. its matching Saliency Heatmap (Right).



Figure 8: The bat "Jafar". The Original image (Left) vs. its matching Saliency Heatmap (Right).

4 Conclusions and Future Steps

Overall, the model demonstrates an ability to recognize individuals from both species. However, in the case of Mauritius bats, it appears to rely more significantly on distinct facial features, suggesting a stronger alignment between visual identity cues and the network's recognition capabilities.

To gain further insight into the features used by the network for recognition, we generated mean saliency maps for both species. These visualizations aggregate the model's attention across multiple samples, offering a coarse approximation of which image regions were generally considered important during inference.

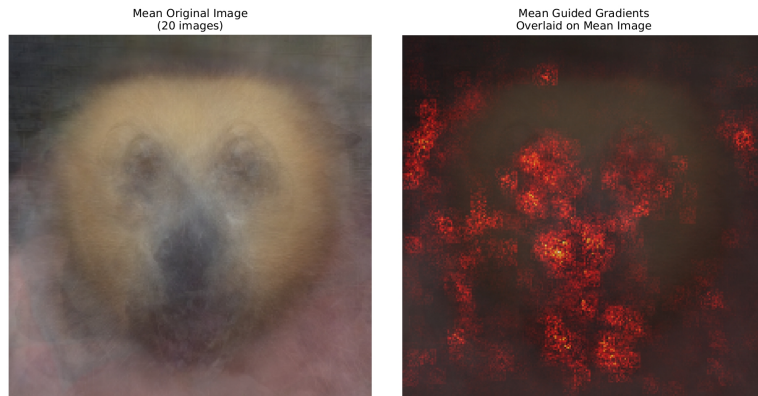


Figure 9: Mean saliency map for Mauritius bats.

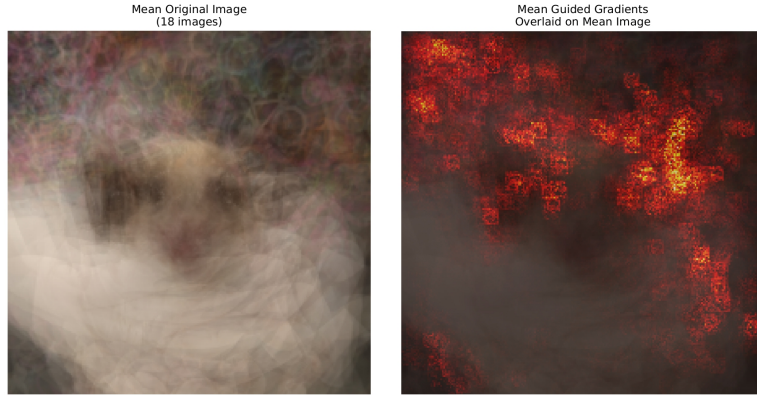


Figure 10: Mean saliency map for *Rousettus* bats.

While preliminary and limited in precision, the saliency map for *Mauritius* bats appears to show stronger emphasis near central facial features. Whereas the saliency for *Rousettus* bats seems more diffusely distributed, roughly aligning with the overall head shape. These observations tentatively support our hypothesis but must be interpreted with caution due to inconsistencies in pose, alignment, and scale across the original input images.

To address these issues and improve interpretability, we plan to refine the saliency map generation process by normalizing facial alignment across images. Specifically, we aim to implement a tool that detects the eyes and nose, then rotates and scales each image such that the triangle formed by these features is consistently oriented—upside down, with the base (eyes) horizontally aligned near the top of the image. If successful, this preprocessing step should better align key facial landmarks across samples, allowing the resulting saliency maps to more reliably indicate consistent regions of model attention.

Additionally, one pattern noted in the current saliency maps is a recurring activation around the outer contour of the head. While it is plausible that facial shape is a meaningful differentiator between individuals, this feature may also reflect a confounding variable unrelated to individual identity. To test this, we intend to conduct a follow-up experiment using circular masks to crop all faces to a fixed outline. This will help isolate the effect of shape-based features and evaluate whether the model can still recognize individuals based on internal facial details alone.

Overall, these next steps aim to clarify whether the model truly learns identity-relevant visual cues or instead relies on incidental patterns in the data. We believe that improved spatial normalization and tighter experimental controls will help resolve some of the current ambiguity.